
The Structure of Spacetime

MASTERS THESIS

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1 Abstract

Using the framework of General Relativity, the simplest universes have been explored using Conformal diagrams. Discussions regard the Minkowski and de Sitter spacetime. The report presumes a working knowledge of general relativity, and presents small elements of cosmology. Parts of Sean Carroll's textbook "Spacetime and Geometry" has been closely followed to provoke discussion, parts of which have been reproduced here to serve as revision.

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2 Introduction

Einstein's theory of General Relativity postulates that spacetime is a 4-dimensional differential manifold with a Lorentzian metric and the Levi-Civita connection, whose energy-mass constituents determine its Ricci curvature tensor. This is encapsulated in the Einstein field equations:

$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4}T_{\mu\nu} \quad (1)$$

where $g_{\mu\nu}$ is the metric tensor, $R_{\mu\nu}$ is the Ricci curvature tensor, R is the Ricci scalar, Λ is the cosmological constant, $T_{\mu\nu}$ is the energy-momentum tensor.

In general, finding exact solutions to these equations, which are components of the metric tensor $g_{\mu\nu}$, poses to be difficult. However, relying on assumptions of symmetries within the system, analytical solutions can be found. The simplest of those solutions are those with the highest degree of symmetry, known as maximally symmetric spaces. These are the solutions which are studied below.

2.1 Cosmology

Cosmological models are based on the cosmological principal which states that on sufficiently large scales, the spatial distribution of matter in the universe is homogeneous and isotropic. Thus, an observer anywhere in the universe will perceive the same physics observing in any direction.

Isotropy and homogeneity imply that a space is maximally symmetric such that the space possess its maximum number of Killing vectors. If a spacetime has a Killing vector, we can find a co-ordinate system in which the metric is independent of one of the co-ordinates. Hence, Killing vectors imply symmetry, and thus conserved quantities associated with the motion of free particles, a quantity that is unchanged in the direction of the Killing vector. On an n -dimensional manifold, the **maximum** number of Killing vectors is

$$\frac{n(n+1)}{2} \quad (2)$$

For a vector V^μ to be a Killing vector, it must satisfy **Killing's equation**

$$\nabla_{(\mu} V_{\nu)} = 0 \quad (3)$$

Recall the definition for the covariant derivative

$$\nabla_\alpha V_\beta = \partial_\alpha V_\beta - \Gamma_{\beta\alpha}^\delta V_\delta \quad (4)$$

Computing the Killing vectors for a general metric is difficult, but given a simple metric, one can guess it's Killing vector and check them via Killing's equation. As an example, the Killing vectors of Minkowski space are derived when discussing the Minkowski space..

Through empirical evidence, we know that space is isotropic and homogeneous but not all of spacetime. However, we study such spacetimes as they provide a theoretical foundation on which we can build upon when adopting a more realistic approach.

Generally, we use the Riemann tensor to encode the curvature of our manifold. The Riemann tensor for a maximally symmetric n -dimensional manifold with metric $g_{\mu\nu}$ can be written as :

$$R_{\rho\sigma\mu\nu} = \kappa(g_{\rho\mu}g_{\sigma\nu} - g_{\rho\nu}g_{\sigma\mu}) \quad (5)$$

where κ is a normalised measure of the Ricci curvature $\kappa = \frac{R}{n(n-1)}$. The Ricci scalar R is constant over the manifold.

At any point on the manifold, we can locally reconstruct the metric to resemble flatspace i.e. $g_{\mu\nu} \sim \eta_{\mu\nu}$ where $\eta_{\mu\nu}$ is the metric for Minkowski space. The types of maximally symmetric manifolds are characterised by the signature of the metric and the sign of the constant κ .

The familiar Minkowski space is the maximally symmetric spacetime with no curvature i.e. $\kappa = 0$.

The maximally symmetric spacetime with positive curvature $\kappa > 0$ is known as **de Sitter space**, which is explored further below.

The spacetime with negative curvature $\kappa < 0$ is the anti-de Sitter space (commonly abbreviated as AdS), which is not discussed here.

How do these spacetimes arise from Einstein's equation?

First, we stick to 4 dimension. By taking the trace of the Riemann tensor as defined in equation (5), ($n = 4$), we arrive at:

$$R_{\mu\nu} = 3\kappa g_{\mu\nu} \quad R = 12\kappa$$

In a maximally symmetric spacetime, we see that the Ricci tensor is proportional to the metric tensor.

Thus the Einstein tensor becomes

$$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} = -3\kappa g_{\mu\nu}$$

Hence, Einstein's equation $G_{\mu\nu} = 8\pi G T_{\mu\nu}$ implies that the energy-momentum tensor is proportional to the metric via

$$T_{\mu\nu} = -\frac{3\kappa}{8\pi G} g_{\mu\nu}$$

A energy momentum tensor of this form corresponds to a vacuum energy (or a cosmological constant). We can model the cosmological constant term in as a stress-energy term arising from the vacuum where the vacuum has a constant positive mass-energy density ρ with constant pressure $p = -\rho$. The relationship between ρ and p (for an ideal gas) as the equation of state.

The energy density and pressure are given by

$$\rho = -p = \frac{3\kappa}{8\pi G}$$

If the energy density is positive ($\rho > 0$) we obtain a de Sitter solution. Similarly, if it's negative we get an anti-de Sitter solution.

Our universe contains ordinary matter and radiation, and possibly vacuum energy. Thus the maximally symmetric spacetimes are not compatible with varying amounts of matter and/or radiation. However, they are useful to study as the represent locally unique solutions to Einsteins equations in the absence of matter or gravitational radiation.

2.2 Robertson Walker Metric

To describe our universe, we must forgo the perfect "cosmological principle" i.e. we no longer assume that spacetime maximally symmetric, but only space i.e. the universe is spatially homogeneous and isotropic while it evolves in time.

In the mathematical context, this translates to the statement that the universe can be constructed from "cross-sectional" spacelike slices such that each slice is maximally symmetric.

Hence, we consider our spacetime to be $\mathbf{R} \times \Sigma$ where $\mathbf{R}$ represents the time direction and Σ is a maximally symmetric three-manifold.

Thus, line element takes the form:

$$ds^2 = dt^2 + R^2(t) d\sigma^2 \quad (6)$$

where t is a time like coordinate, $R(t)$ is the **scale factor** which indicates how big Σ is at time t and $d\sigma^2$ is the metric on Σ .

$d\sigma^2$ can be expressed as

$$d\sigma^2 = \gamma_{\mu\nu}(u) du^i du^j$$

where u^a are the coordinates on Σ and $\gamma_{\mu\nu}$ is a three dimensional maximally symmetric metric. We use **comoving coordinates*** to express the metric, which alleviates us of off-diagonal terms.

Comoving coordinates assign constant spatial values to observers who perceive the universe as isotropic. Comoving observers are called so as they move *with* the Hubble flow: the motion of astronomical objects due to the expansion of the universe only.

As we require the space three-metric to be γ_{ij} to be maximally symmetric, we know it is spherically symmetric. Thus, we can re-express the metric using spherical coordinates (following the Schwarzschild solution)

$$d\sigma^2 = \frac{d\bar{r}^2}{1 - k\bar{r}^2} + \bar{r}^2 d\Omega^2 \quad (7)$$

where $k = R/6$ is the curvature, dependent on the Ricci scalar curvature of the spatial line element, $\bar{r}$ is the radial coordinate and $d\Omega^2$ is the metric on a two-sphere. k also determines the size of the spatial surfaces via the Ricci tensor. We normalise the curvature k such that k lies in $[-1, 0, 1]$.

Introducing the radial coordinate $d\chi = \frac{d\bar{r}}{\sqrt{1-k\bar{r}^2}}$, allows use to re-write the metric as

$$d\sigma^2 = d\chi^2 + S_k^2(\chi) d\Omega^2 \quad (8)$$

where the solutions for $\bar{r} = S_k(\chi)$ are obtained by integration:

$$S_k \equiv \begin{cases} \sin(\chi), & k = +1 \\ \chi, & k = 0 \\ \sinh \chi, & k = -1 \end{cases} \quad (9)$$

We recognise the case $k = 0$ corresponding to no curvature i.e **flat** Euclidean space.

The **closed** case $k = +1$ corresponds to positive curvature, as the metric becomes the metric on the three-sphere.

The **open** case $k = -1$ corresponds to negative curvature on Σ .

With the knowledge of the form of the spatial metric (7), the metric (6) becomes

$$ds^2 = -dt^2 + R^2(t) \left[\frac{d\bar{r}^2}{1-k\bar{r}^2} + \bar{r}^2 d\Omega^2 \right] \quad (10)$$

This is the **Robertson Walker** metric. It is common to re-express this metric with more convenient parameters. By using the invariance of the coordinates to particular coordinate transformations, we use the **dimensional scale factor** $a(t) = R(t)/R_0$, distance coordinate $r = R_0\bar{r}$ and curvature $\kappa = k/R_0^2$. Now, κ can range any value. Using these parameters instead, we arrive at the Robertson Walker metric in its familiar form

$$ds^2 = -dt^2 + a^2(t) \left[\frac{dr^2}{1-\kappa r^2} + r^2 d\Omega^2 \right] \quad (11)$$

The Ricci scalar is then:

$$R = 6 \left[\frac{\ddot{a}}{a} + \left(\frac{\dot{a}}{a} \right)^2 + \frac{\kappa}{a^2} \right] \quad (12)$$

2.3 The Friedmann Equations

The Robertson Walker metric is defined for any behaviour of the scale factor $a(t)$. Using Einstein's equations, we can derive the Friedmann equations which relate the scale-factor to the energy-momentum of the universe. By choosing to model matter and energy by a perfect fluid, we can separate the forms of the energy density ρ to radiation, matter and vacuum dominated.

The Friedmann equations are as follows::

$$\left(\frac{\dot{a}}{a} \right)^2 = \frac{8\pi G}{3} \rho - \frac{\kappa}{a^2} \quad (13)$$

and

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}(\rho + 3p) \quad (14)$$

In a vacuum dominated universe, the energy density is constant, $\rho_\Lambda \propto a^0$. As the universe expands, the energy density in matter and radiation decreases. In the presence of a non-zero vacuum energy, the universe becomes **vacuum-dominated**. de Sitter and anti-de Sitter are vacuum-dominated solutions.

The universe consists of numerous types of energy densities. However, as the different sources evolve at different rates, considering long duration, a single source will dominate the contribution to the energy density. Thus we it is useful to examine the solutions to the Friedmann equations with one type of energy density $\rho \propto a^{-n}$

As the spatial curvature contributes as an energy source, we are considering flat universes dominated by a single source or empty universes with spatial curvature.

Thus the Friedmann equation implies

$$\dot{a} = a^{1-n/2}$$

which results in

$$a \propto t^{2/n} \quad (15)$$

2.4 What are Conformal Diagrams?

Conformal diagrams (aka Penrose diagrams) allow us to visualise an infinite curved spacetime manifolds. They are informative of the global properties and causal structure. Thus, given two different universes and their conformal diagrams, we can tell them apart at glance. To create conformal diagrams, we need to manipulate the metric via coordinate transformations such that the resulting metric is conformally related to the initial through a **conformal transformation**.

2.4.1 Conformal Transformations

The metric $g_{\mu\nu}$ is conformally related to $\tilde{g}_{\mu\nu}$ if

$$\tilde{g}_{\mu\nu} = \omega^2(\mathbf{x}) g_{\mu\nu}$$

with the equivalent being

$$\tilde{ds}^2 = \omega^2(\mathbf{x}) ds^2$$

for some non-vanishing function $\omega(\mathbf{x})$, where $\mathbf{x}$ is used to denote the spacetime coordinates x^μ .

A conformal transformation is not just a change of coordinates, but a change of geometry. Thus, the geodesics of the two metrics generally differ. However, conformal transformations leave null curves invariant. This follows from the definition of a null geodesic.

If a curve is null with respect to $g_{\mu\nu}$, it is also null to the conformally related metric $\tilde{g}_{\mu\nu}$. Recall that a curve is null if its tangent vector $\frac{dx^\mu}{d\lambda}$ is null, i.e.

$$g_{\mu\nu} \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda} = 0$$

Hence, it follows that :

$$\tilde{g}_{\mu\nu} \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda} = \omega^2(x) g_{\mu\nu} \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\lambda} = 0$$

2.4.2 Conformal Diagrams

For the diagram to be of use, we would like a time-like coordinate T and a radial one R , with the feature that light ones can be portrayed at 45° , or $dT/dR = \pm 1$ in natural units i.e. $c = 1$. This is so we can observe the causal structure.

Consider a metric in a spherical symmetric coordinate system, with a timelike coordinate t and spacelike coordinate r , with infinite or semi-infinite ranges.

We require finite coordinate ranges in order to represent the system in a conformal diagram. Thus, we make the following sequence of coordinate transforms.

Firstly, we construct **null coordinates** by summing the radial and timelike coordinate:

$$u = t - r \quad v = t + r \quad (16)$$

Then, we restrict their range of values to bring them to a finite interval by an appropriate function, for example:

$$U = \arctan u \quad V = \arctan v \quad (17)$$

Then, we reconstruct our timelike and spacelike coordinate by taking sums:

$$T = V + U \quad R = V - U \quad (18)$$

We can use these coordinates to construct conformal diagrams with their finite infinite range. The usefulness of this methodology is presented when naively constructing the spacetime diagram of Minkowski space.

3 Minkowski Space

The Minkowski space is a 4D manifold absent of curvature, and hence the playground for special relativity. It's defined by its metric $\eta_{\mu\nu} = \text{diag}(-1, 1, 1, 1)$. In polar coordinates, the line element becomes

$$ds^2 = -dt^2 + dr^2 + r^2 d\Omega^2 \quad (19)$$

where $d\Omega^2 = d\theta^2 + \sin^2\theta d\phi^2$ is the metric on a unit two-sphere. Null rays being $t = \pm r$ with the ranges of the coordinates being

$$-\infty < t < \infty, \quad 0 \leq r < \infty \quad (20)$$

Given the infinite ranges, it would not be possible to illustrate this spacetime in complete. Thus, we try to obtain finite ranges with co-ordinate transformations.

3.1 Killing vectors of Minkowski space

Consider the Minkowski spacetime in Cartesian coordinates (t, x, y, z) with line element $ds^2 = -dt^2 + dx^2 + dy^2 + dz^2$

Using our knowledge of Minkowski space from special relativity, we can guess the form of Killing vectors.

Firstly, note that the metric components are independent of the coordinates themselves i.e. (t, x, y, z) . Therefore, we can presume that the translations in any one of the axes would correspond to a symmetry of the metric. Hence, we can guess that these symmetries are generated by the four Killing vectors:

$$T^\mu = (1, 0, 0, 0)$$

$$X^\mu = (0, 1, 0, 0)$$

$$Y^\mu = (0, 0, 1, 0)$$

$$Z^\mu = (0, 0, 0, 1)$$

To see that these satisfy Killing's equation (3), recall the covariant derivative definition (4). We know that for Minkowski space, the connection coefficients vanish and thus the covariant derivative is simply a partial derivative i.e. $\nabla_\mu K_\nu = \partial_\mu K_\nu$. Now, looking at the Killing vectors' components, we note that they are constant and thus the partial derivatives also vanish. Hence, Killing's equation is satisfied for these vectors.

Now we can consider rotations around the spatial axes. Consider the 3D cartesian coordinate axes. Implementing an anti-clockwise rotation about the x-axis, and observing the components lead to the rotation vector $R_{x_\mu} = (0, 0, -z, y)$. Killing's condition for our rotation vector $\nabla_\mu R_{x_\nu} = 0$, boils down to looking at the two cases $\mu = \nu$ and $\mu \neq \nu$

For the former case, it is clear that the components of the rotation is independent of the μ^{th} coordinate and hence $\nabla_\nu R_{x_\nu} = 0$. For the latter case, the only non-trivial case is

$$\nabla_y R_{x_z} + \nabla_z R_{x_y} = \partial_y(y) + \partial_z(-z) = 0 \quad (21)$$

Hence, Killing's equation is satisfied. A similar procedure leads to rotations around the y and z axis $R_{y_\mu} = (0, z, 0, -x)$ and $R_{z_\mu} = (0, -y, x, 0)$. After raising the index using the metric $\eta_{\mu\nu} = \text{diag}(-1, 1, 1, 1)$, we arrive at the Killing vectors associated with our spatial rotations:

$$\begin{aligned} R_{x^\mu} &= (0, 0, -z, y) \\ R_{y^\mu} &= (0, z, 0, -x) \\ R_{z^\mu} &= (0, -y, x, 0) \end{aligned}$$

We can guess that a boost in each axis maybe Killing vectors, and it would a correct guess. Checking the Boost Killing vectors (listed below) follow the same arguments as the rotation vectors.

$$\begin{aligned} B_{x^\mu} &= (x, t, 0, 0) \\ B_{y^\mu} &= (y, 0, t, 0) \\ B_{z^\mu} &= (z, 0, 0, t) \end{aligned}$$

We arrive at a total of 10 Killing vectors for the Minkowski space. The maximum number of Killing vectors for a 4D Minkowski spacetime is 10, as calculated by (2) with $n = 4$. Hence, the Minkowski spacetime is a maximally symmetric.

3.2 The Process

First, we illustrate the usefulness of using null coordinates by naively approaching the problem of bounding our coordinates. The arctangent function maps the real line to a finite interval, hence may useful. Thus, we try defining $\bar{t} = \arctan t$ and $\bar{r} = \arctan r$ with ranges

$$-\pi/2 < \bar{t} < \pi/2, \quad -\pi/2 < \bar{r} < \pi/2$$

At this point, we can note that the slope of the light cones are given by $d\bar{t}/d\bar{r} = \pm \cos^2 \bar{t} / \cos^2 \bar{r}$ which is not equal to ± 1 as required. Using these coordinates in a spacetime diagram, lead to unclear null geodesics, making it difficult to observe the causal structure within the spacetime.

By separating the variables and integrating, we reach $\tan \bar{t} = \tan \bar{r} + C$, where C is a constant of integration. By varying this constant, we achieve the family of light-like curves. Figure 1 illustrates this procedure.

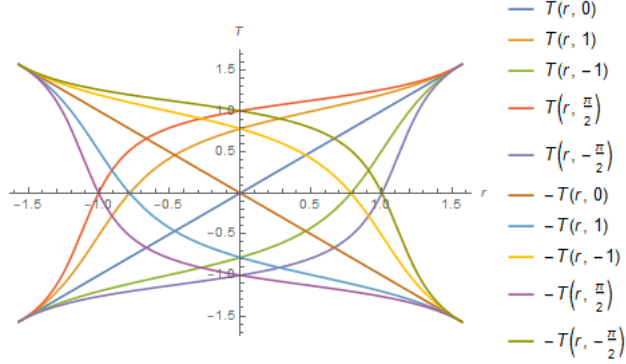


Figure 1: A diagram for Minkowski spacetime after the co-ordinate transformation $\bar{t} = \arctan t$ and $\bar{r} = \arctan r$. A function $T(r, c)$ returns the curve T given $\bar{r}$ and constant C , i.e. the null ray. Null rays with different constants are shown. We can observe that not all null rays not travel at 45° , hence making this coordinate transformation inappropriate for conformal diagrams.

The null rays, displayed in the figure above (1), do not have a constant gradient of ± 1 as we wished. The approach we should take are the coordinate transformations outlined by equations (12) - (14), making use of the null coordinates. This procedure is fully demonstrated in Carroll's text [1]. We illustrate this procedure for the case of the expanding universe instead.

Following the coordinate transformations, we can arrive at the metric

$$ds^2 = \omega^{-2}(T, R)(-dT^2 + dR^2 + \sin^2 R d\Omega^2) \quad (22)$$

where the conformal factor ω

$$\omega = \cos T + \cos R \quad (23)$$

The coordinate ranges for our timelike coordinate T and R become

$$0 \leq R < \pi, \quad |T| + R < \pi \quad (24)$$

Hence, the original Minkowski metric (19) can be thought to be related to this metric by a conformal transformation

$$\tilde{ds}^2 = \omega^2(T, R) ds^2 \quad (25)$$

$$= -dT^2 + dR^2 + \sin^2 R d\Omega^2 \quad (26)$$

This metric describes the manifold $\mathbf{R} \times S^3$, where the 3-sphere is spacelike, constant in time. Though we've introduced curvature into the metric, we need not worry as it is nonphysical and arises due to our choice of coordinates.

The metric we've arrived to is that of the "Einstein Static Universe", a solution to Einstein's equation with a perfect fluid and a cosmological constant.

3.2.1 The Penrose Diagram

Figure 2 below is the conformal diagram for Minkowski spacetime. The space-time is mapped onto the subspace defined by (20), as opposed to the full range defined by (16).

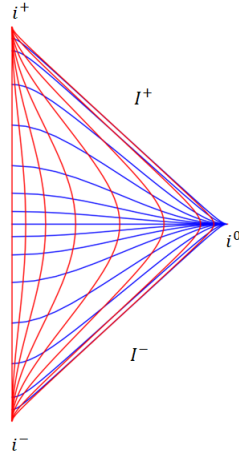


Figure 2: Conformal Diagram for Minkowski space in coordinates R and T . Curves of constant r and t are shown in red and blue respectively. Each point on the diagram represents a 2-sphere, except for i^- , i^0 , i^+ which are *actually* points. I^+ and I^- are null surfaces with topology of $\mathbf{R} \times S^2$

From the co-ordinates above,

$$i^+ = (T = \pi, R = 0)$$

$$i^0 = (T = 0, R = \pi)$$

$$i^- = (T = -\pi, R = 0)$$

$$I^+ = (T = \pi - R, 0 < R < \pi)$$

$$I^- = (T = \pi + R, 0 < R < \pi)$$

There are a few important points to note about the diagram. Each point in the diagram represents a two-sphere, except for a few special points/edges. The radial

null geodesics are at $\pm 45^\circ$ in the diagram, as we wished to preserve the causal structure of the spacetime.

All timelike geodesics begin at past timelike infinity i^- and end at future timelike infinity i^+ , which are *actually* points. All null geodesics begin at past null infinity I^- and end at future null infinity I^+ , where I^- and I^+ are null surfaces. All spacelike geodesics begin and end at spatial infinity i^0 .

The entire Minkowski space has been represented in this figure. From this diagram, it is easy to see that Minkowski space is completely casually connected. Any observer's past and future light-cone will contain the entire spacetime, hence all events within the space will be observable given enough time. The absence of curvature within the spacetime results in this simple geometry.

4 The Expanding Universe

Consider the expanding universe. Using polar coordinates, the metric for this space becomes

$$ds^2 = -dt^2 + t^{2q}(dr^2 + r^2 d\Omega^2) \quad (27)$$

where $d\Omega^2$ is the line element on a two-sphere, and the scale factor follows a power law $a(t) = t^q$ with $0 < q < 1$. Note that this metric contains a singularity at $t = 0$, which then restrict our coordinates to:

$$0 < t < \infty, \quad 0 \leq r < \infty \quad (28)$$

We can conformally relate the metric to Minkowski space.

We specify the new time coordinate η as **conformal time**, which satisfies

$$\eta = \frac{1}{1-q} t^{1-q} \quad (29)$$

from which we have $d\eta = t^{-q} dt \rightarrow dt^2 = t^{2q} d\eta^2$. Note that the range for η is the same as the range for t .

Substituting η for t in the metric:

$$\begin{aligned} ds^2 &= -dt^2 + t^{2q} [dr^2 + r^2 d\Omega^2] \\ &= -t^{2q} d\eta^2 + t^{2q} [dr^2 + r^2 d\Omega^2] \\ &= t^{2q} [-d\eta^2 + dr^2 + r^2 d\Omega^2] \\ &= (\eta(1-q))^{\frac{2q}{1-q}} [-d\eta^2 + dr^2 + r^2 d\Omega^2] \end{aligned}$$

Now, our expanding universe metric is the product of the Minkowski metric and a conformal factor. Hence, for the conformal diagram, we can repeat our coordinate transformation equations (12)-(14).

Firstly, we define null coordinates.

$$u = \eta - r, \quad v = \eta + r \quad (30)$$

$$-\infty < u < \infty, \quad -\infty < v < \infty, \quad u \leq v \quad (31)$$

From above we have $\eta = \frac{1}{2}(v+u)$ and $r = \frac{1}{2}(v-u)$.

Substituting for η and r into the metric, we arrive at

$$ds^2 = \left(\frac{1}{2}(v+u)(1-q) \right) \left[-\frac{1}{2}[dv du + du dv] + \frac{1}{4}(v-u)^2 d\Omega^2 \right]$$

Now, we can use the arctangent function to restrict the ranges of u and v .

$$U = \arctan u, \quad V = \arctan v \quad (32)$$

$$-\pi/2 < U < \pi/2, \quad -\pi/2 < V < \pi/2, \quad U \leq V \quad (33)$$

Again, substituting for u and v leads to the metric:

$$ds^2 = \left(\frac{(\sin(U+V)(1-q))^{\frac{2q}{1-q}}}{2\cos V \cos U} \right) \frac{[-2(dV dU + dU dV) + \sin^2(V-U)d\Omega^2]}{4\cos^2 U \cos^2 V}$$

Now, going back to our timelike and spacelike coordinates T and R ,

$$T = V + U, \quad R = V - U \quad (34)$$

$$0 \leq R, \quad T + R < \pi \quad (35)$$

Substituting for these into our metric above:

$$ds^2 = \omega^{-2}(T, R) \left[-dT^2 + dR^2 + \sin^2 R d\Omega^2 \right] \quad (36)$$

where

$$\omega = (\cos T + \cos R) \left[\frac{\cos T + \cos R}{\sin T(1-q)} \right]^{\frac{q}{1-q}} \quad (37)$$

By following a similar procedure to that of Minkowski space, we now have our metric of the expanding universe conformally related to the Einstein Static Universe. There is, however, a difference between the cases which is that the timelike coordinate halts at a singularity at $T = 0$.

This conformal diagram is the upper half of the conformal diagram for Minkowski space. However, we can choose two events in the spacetime with the property that their past lightcones will hit the singularity before the intersect, hence are causally disconnected.

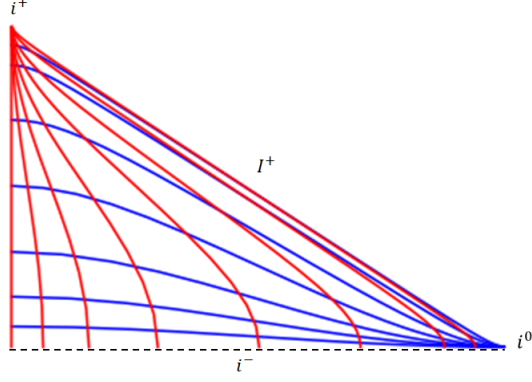


Figure 3: Conformal Diagram for the Robertson Walker universe with $a(t) \propto t^q$ for $0 < q < 1$. The dashed line at the bottom edge represents the singularity at $t = 0$ ($T = 0$). Curves of constant r and t are shown in red and blue respectively.

5 Milne Universe

Consider the completely empty universe with energy density $\rho = 0$, but with spatial curvature. The Freidmann equations (13, 14) become

$$\dot{a}^2 = -k, \quad \ddot{a} = 0 \quad (38)$$

and thus we have that $k = -1$. The solution for the scale factor thus is $a \propto t$. The metric of the Milne universe (via the Robertson Walker metric of curvature $k = -1$) therefore has the form

$$ds^2 = -dt^2 + t^2[dr^2 + \sinh^2 r d\Omega^2] \quad (39)$$

Using the expression for scalar curvature in Robertson-Walker metric, we get $R = 0$ as we expect for a matterless spacetime.

If we recall the metric for Minkowski spacetime in conformal coordinates (T, R, θ, ϕ) and make the coordinate transformation to $(\bar{t}, r, \theta, \phi)$ via $t = \bar{t} \cosh r$ and $r = \bar{t} \sinh r$, we arrive at the metric for the Milne universe. Given these coordinates, we have $t^2 - r^2 > 0$ and hence $t > |r|$, with the equality for null rays. [4]

Thus we realise that the Milne universe is a re-parameterised patch of Minkowski space. The spatial slices with $T = \text{constant}$ are flat, while the slices $\bar{t} = \text{constant}$ are hyperboloids. These have negative scalar curvature.

To confirm that they do indeed represent the same spacetime, it is possible to compute all the components of the Riemann tensor and verify that they vanish, and as we know - any spacetime with vanishing Riemann curvature is locally Minkowski.

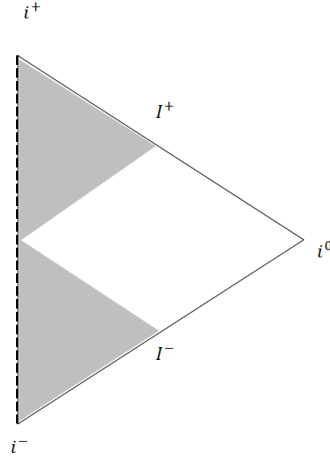


Figure 4: The Milne universe as a patch on Minkowski spacetime. The shaded grey area represents the Milne universe.

6 de Sitter Space

The maximally symmetric spacetime with positive curvature $\kappa > 0$ is called **de Sitter space**. The n -dimensional de Sitter space dS_n may be realised as the hypersurface described by the equation $-X_0^2 + X_1^2 + \dots + X_n^2 = \alpha^2$ where α is the de Sitter radius. The radius is related to the curvature Ricci scalar by $R_{dS_n} = n(n-1)/\alpha^2$.

A way to understand this space is to consider a five-dimensional Minkowski space with the metric

$$ds_5^2 = -du^2 + dx^2 + dy^2 + dz^2 + dw^2$$

We can embed a 4D hyperboloid in this space given by

$$-u^2 + x^2 + y^2 + z^2 + w^2 = \alpha^2$$

Inducing coordinates onto hyperboloid allows us to re-express the metric in a familiar form. The coordinates we choose are $[t, \chi, \theta, \phi]$ via the equations:

$$\begin{aligned} u &= \alpha \sinh(t/\alpha) \\ w &= \alpha \cosh(t/\alpha) \\ x &= \alpha \cosh(t/\alpha) \sin \chi \cos \theta \\ y &= \alpha \cosh(t/\alpha) \sin \chi \sin \theta \cos \phi \\ z &= \alpha \cosh(t/\alpha) \sin \chi \sin \theta \sin \phi \end{aligned}$$

The metric becomes

$$ds^2 = -dt^2 + a^2 \cosh^2(t/\alpha) \left[d\chi^2 + \sin^2 \chi (d\theta^2 + \sin^2 \theta d\phi^2) \right] \quad (40)$$

The ranges of our induced coordinates are as follows:

$$\begin{aligned} -\infty < t < \infty \\ 0 \leq \chi \leq \pi \\ 0 \leq \theta \leq \pi \\ 0 \leq \phi \leq 2\pi \end{aligned}$$

Within the metric (40), we recognise the expression in the round parentheses as the line element on a two-sphere $d\Omega_2^2$, and the expression within the square brackets is the line element on a three-sphere $d\Omega_3^2$. Apart from the coordinate singularities at $\chi = 0, \pi$, $\theta = 0, \pi$, and $\phi = 0, 2\pi$, we have a complete map of dS_4 and hence the coordinates of choice are referred to as the global coordinates. We can deduce that dS_4 is spatially closed. Using this coordinate system, we can interpret de Sitter space as describing a spatial three-sphere which initially shrinks to a minimum "radius" at $t = 0$, before re-expanding.

By suppressing two dimensions, we can create a visual representation of the space as a 2-dimensional hyperboloid inside 3-dimensional space.

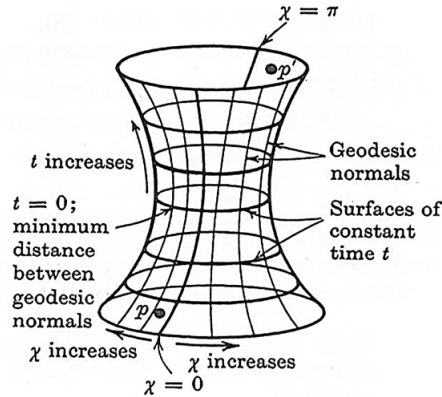


Figure 5: de Sitter space-time represented by a hyperboloid embedded in a five-dimensional flat space (2 dimensions suppressed). The coordinates $[t, \chi, \theta, \phi]$ cover the whole hyperboloid; the sections $t = \text{constant}$ are surfaces of curvature $k = +1$. Figure taken from [2]

Global coordinates are the **closed slicing** of de Sitter. Recall that the FRW metric for any homogenous and isotropic universe is $ds^2 = -dt^2 + a^2(t)d\Sigma^2$ where the spatial slices $d\Sigma$ are either open (hyperbola), flat or closed (spheres). [3]

de Sitter is an example of a closed FRW universe. The scale factor $a(t) = \alpha \cosh(t/\alpha)$ is exponentially decreasing, reaches a minimum at $t = 0, a(0) = \alpha$ then exponentially expands into the future.

The topology of de Sitter is $\mathbf{R} \times S^3$ and thus it is trivial to derive the conformal diagram, as the Einstein Static universe is also a spacetime of topology $\mathbf{R} \times S^3$. To conformally relate the two metric, we make the coordinate transformation from t to t' via

$$\cosh(t/\alpha) = \frac{1}{\cos(t')} \quad (41)$$

Hence, the metric becomes

$$ds^2 = \frac{\alpha^2}{\cos^2(t')} d\bar{s}^2 \quad (42)$$

where $d\bar{s}^2$ is the metric for the Einstein static universe.

The range of the new time coordinate is $-\pi/2 < t' < \pi/2$

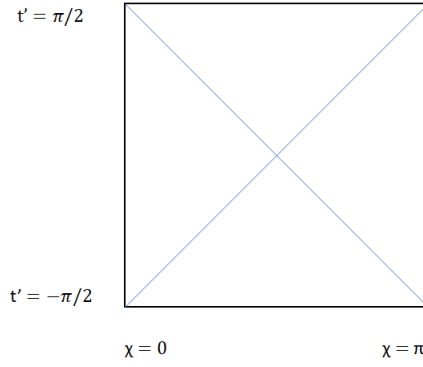


Figure 6: The conformal diagram of de Sitter space-time. A spacelike slice of constant t' represents a three-sphere.

On the conformal diagram above, the vertical edges on the left and right of the square are the north and south pole of the sphere respectively. The diagonal lines represent null rays: a photon released at past infinity will reach the antipodal point on the sphere at future infinity.

There are points on de Sitter space which cannot be connected to each other by any geodesic. We can have two points which are causally disconnected i.e. whose future (or past) light cones do not intersect. This can be attributed to the fact that the spherical spatial sections are expanding so rapidly that light from one point can never come into contact with light from the other.

In contrast to Minkowski space, de Sitter space has a spacelike infinity for time-like and null lines, both in the future and the past. Thus all observers approach these boundaries in the infinite past and future given infinite-affine time to reach them.

7 Conclusion

We've explored maximally symmetric spaces and then FRW universes. Using conformal diagrams, it is easy to study the causal structure of the spacetime and build a geometric understanding of the space. By studying the diagrams, we are lead to surprising connections between spacetimes.

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